When models answer “No,” do they give a valid counterexample or the smallest repair?

These are examples demonstrating that LLMs can flag an error and either provide a repair or give a counterexample.

QID	Model	Evidence flagged	Core move
Q20	GPT-5 high	Repair	Halve the exponential-mechanism exponent
Q27	Claude Opus	Repair	Restore norm clipping with $\min$ and $\|g\|_2$
Q71	Gemini 3	Repair	Restrict the TNB parameter to $\gamma \in (0, 1)$
Q67	GPT-5 high	Counterexample	0-RDP trainers plus a private $\arg\max$ reveal a bit
Q120	GPT-5 high	Counterexample	Randomized response violates the proposed RDP bound
Q128	GPT-5 high	Counterexample	Neighboring inputs induce $D_2 = \infty$

EXAMPLE 1 · REPAIR

Q20 · DP selection: expoMech

QUESTION	Q20	MODEL	GPT-5 high · run 764
TOPIC	DP selection: expoMech	NEGATIVE MODE	`conclusion_perturbation`
SOURCE	`dp-judge_gpt-5-high_cot_764_responses.json`		
EVIDENCE	Repair		

Typeset question (from `cate_2.json`)

The pseudocode below implements the exponential mechanism. Does this implementation ensure ε-differential privacy under the replace-one neighboring relationship?

Algorithm 1: Exponential Mechanism

Input : Dataset $D = \{z_i\}_{i=1}^n$; candidate set Y; utility (score) function $q(D, y) = \sum_{i=1}^n 1\{z_i = y\}$; privacy parameter ε
Output: Selected item $\hat{y} \in Y$ sampled by the exponential mechanism

for $y \in Y$ do
  $p_y \leftarrow \frac{\exp(\varepsilon\, q(D,y))}{\sum_{y' \in Y} \exp(\varepsilon\, q(D,y'))}$
Sample $\hat{y} \sim \text{Multinomial}(1; \{p_y\}_{y \in Y})$
return $\hat{y}$

Selected model response (wording preserved; mathematics typeset)

Explanation/derivation.

- The exponential mechanism with score function q and global sensitivity Δq selects y with probability proportional to
$$\exp\left(\frac{\varepsilon q(D, y)}{2\Delta q}\right)$$
to achieve ε-differential privacy.
- Here,
$$q(D, y) = \sum_{i=1}^{n} 1\{z_i = y\}$$
is a count. Under replace-one adjacency, for any fixed y we have $|q(D, y) - q(D', y)| \le 1$, so $\Delta q = 1$.
- The provided code uses $p_D(y) \propto \exp(\varepsilon q(D, y))$; that is, it omits the factor $1/2$.

Privacy calculation. Let
$$Z(D) = \sum_{y'} \exp(\varepsilon q(D, y')).$$
Then for neighboring D, D',
$$\frac{p_D(y)}{p_{D'}(y)} = \exp(\varepsilon(q(D, y) - q(D', y))) \frac{Z(D')}{Z(D)}.$$
The first factor is at most e^{ε}. Since $q(D', y') \le q(D, y') + 1$ for every y',
$$Z(D') \le e^{\varepsilon} Z(D).$$
Hence $p_D(y)/p_{D'}(y) \le e^{2\varepsilon}$, and symmetrically for the reverse ratio. The mechanism is 2ε-DP, not ε-DP.

To achieve ε-DP, use

$$p_y \propto \exp\Big(\frac{\varepsilon}{2} q(D, y)\Big),$$

equivalently replacing ε in the code by $\varepsilon/2$.

No

FLAG · REPAIR

"To achieve ε-DP, use $p_y \propto \exp((\varepsilon/2)q(D, y))$ (equivalently, replace ε in the code by $\varepsilon/2$)."

COMMENTS FROM DATASET

The exponent in the density of the exponential mechanism is missing a factor of $1/2$. Consequently, under the replace-one neighboring relation, it satisfies only 2ε-DP.

EXAMPLE 2 · REPAIR

Q27 · DP-GD

QUESTION	Q27	MODEL	Claude Opus · run 89
TOPIC	DP-GD	NEGATIVE MODE	algorithm_perturbation
SOURCE	dp-judge_claude-opus_cot_89_responses.json		
EVIDENCE	Repair		

Typeset question (from `cate_2.json`)

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy μ-Gaussian differential privacy under the add/remove neighboring relationship (assume number of samples n is public information)?

Algorithm 2: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T; step sizes $\{\eta_t\}_{t=0}^{T-1}$; clipping norm $C > 0$; target gaussian DP μ

Output: Final parameter θ_T

for $t = 0, \dots, T-1$ do
  for $i = 1, \dots, n$ do
    $g_i \leftarrow \nabla_\theta \ell(\theta_t; z_i)$
    $\hat{g}_i \leftarrow g_i \cdot \max\left(1, \frac{C}{\|g_i\|_2^2}\right)$
  $\tilde{g}_t \leftarrow \frac{1}{n}\left(\sum_{i=1}^n \hat{g}_i + \frac{C\sqrt{T}}{\mu}\xi_t\right), \quad \xi_t \sim \mathcal{N}(0, I_d)$
  $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{g}_t$
return θ_T

Selected model response (wording preserved; mathematics typeset)

Issue 1: Incorrect Clipping Operation. The clipping operation in line 4 appears to be incorrect. The standard gradient clipping should be

$$\hat{g}_i \leftarrow g_i \cdot \min\left(1, \frac{C}{\|g_i\|_2}\right).$$

Instead, the algorithm uses

$$\max\left(1, \frac{C}{\|g_i\|_2^2}\right),$$

which has a squared norm in the denominator, uses max instead of min, and would amplify gradients when $\|g_i\|_2^2 < C$ instead of clipping them. This means gradients are not properly bounded, which is essential for differential privacy.

[...]

Conclusion. Despite the noise calibration appearing correct, the fundamental issue is that the clipping operation is incorrectly implemented. Without proper gradient clipping, the sensitivity is unbounded, and no finite fixed noise scale can guarantee the claimed Gaussian differential privacy. The algorithm fails to satisfy μ-Gaussian differential privacy.

No

FLAG · REPAIR

"The standard gradient clipping should be $\hat{g}_i \leftarrow g_i \cdot \min(1, C/\|g_i\|_2)$."

COMMENTS FROM DATASET

The gradient clipping is implemented incorrectly: the normalization term should be $\|g_i\|_2$, not $\|g_i\|_2^2$. Thus, the stated privacy guarantee doesn't hold in general.

EXAMPLE 3 · REPAIR

Q71 · Hyperparameter-tuning

QUESTION	Q71	MODEL	Gemini 3 · run 17
TOPIC	Hyperparameter-tuning	NEGATIVE MODE	assumption_perturbation
SOURCE	dp-judge_gemini-3_cot_17_responses.json		
EVIDENCE	Repair		

Typeset question (from `cate_2.json`)

Let $\mathrm{TNB}_{\eta,\gamma}$ denote the truncated Negative Binomial distribution: with parameters $\eta \in (-1,\infty)$ and $\gamma > 0$, supported on positive integers. For $\eta \neq 0$, $\Pr[K=k] = \frac{(1-\gamma)^k}{\gamma^{-\eta}-1}\prod_{\ell=0}^{k-1}\frac{\ell+\eta}{\ell+1}$ and $\mathbb{E}[K] = \frac{\eta(1-\gamma)}{\gamma(1-\gamma^{\eta})}$. For $\eta = 0$, $\Pr[K=k] = \frac{(1-\gamma)^k}{k\log(1/\gamma)}$ and $\mathbb{E}[K] = \frac{1/\gamma-1}{\log(1/\gamma)}$. If $\mathcal{D} = \mathrm{TNB}_{\gamma,3}$ with $\gamma = 3$ and each M_j satisfies (α,ε)-RDP with $\varepsilon > 1$ for all $\alpha > 1$, does the algorithm satisfy (α,ε')-RDP with $\varepsilon' = (2-\frac{1}{\alpha})\varepsilon + \frac{\log(1/\gamma)}{\alpha} + \frac{\log(5-\gamma)-\log\gamma-\log(\log(1/\gamma))}{\alpha-1}$?

Algorithm 3: Private Hyperparameter Tuning

Input : Dataset D; DP base training mechanisms $\{M_j\}_{j=1}^{m}$ (one per hyperparameter setting j); distribution for number of runs $\mathcal{D}$; validation metric score$(\cdot)$ computed on a public holdout set
Output: Best model $\hat{\theta}$ and chosen hyperparameter $j^\star$

Draw $K \sim \mathcal{D}$
for $k \leftarrow 1, 2, \ldots, K$ do
  Sample $j_k \sim \mathrm{Unif}(\{1,\ldots,m\})$
  $\theta_k \leftarrow M_{j_k}(D)$
  $s_k \leftarrow \mathrm{score}(\theta_k)$
$k^\star \leftarrow \arg\max_{k\in\{1,\ldots,K\}} s_k$
$j^\star \leftarrow j_{k^\star}, \quad \hat{\theta} \leftarrow \theta_{k^\star}$
return $(\hat{\theta}, j^\star)$

Selected model response (wording preserved; mathematics typeset)

To determine whether the algorithm satisfies the stated (α,ε')-RDP guarantee, we must first check the validity of the distribution $\mathcal{D}$ and the bound at $\gamma = 3$.
1. Validity of the distribution. The probability mass function for $\eta \neq 0$ is

$$\Pr[K=k] = \frac{(1-\gamma)^k}{\gamma^{-\eta}-1}\prod_{\ell=0}^{k-1}\frac{\ell+\eta}{\ell+1}.$$

Substituting $\eta = 3$ and the second parameter $\gamma = 3$ gives

$$\Pr[K=k] \propto (1-3)^k\binom{k+2}{k} = (-2)^k\binom{k+2}{k}.$$

The factor $(-2)^k$ alternates sign, but probability mass must be nonnegative. Thus $\mathrm{TNB}_{3,3}$ does not define a valid probability distribution. For the supplied formula one requires $\gamma \in (0,1)$, which makes $1-\gamma > 0$ and yields a convergent series.
2. Validity of the RDP expression. At $\gamma = 3$,

$$\log(1/\gamma) = \log(1/3) = -\log 3 < 0,$$

so $\log(\log(1/\gamma))$ is not real. Consequently ε' is undefined as a real RDP parameter.
Conclusion. The distribution is ill-defined at $\gamma = 3$ and the bound is not real, so the algorithm cannot satisfy the stated guarantee.

No

FLAG · REPAIR

"For the TNB distribution to be valid using the provided formula, one requires $\gamma \in (0, 1)$, ensuring $1 - \gamma > 0$ and convergence of the series."

COMMENTS FROM DATASET

To define a valid probability distribution, γ must lie in $[0, 1]$; otherwise the resulting density can take negative values. Therefore, the statement is false.

EXAMPLE 4 · COUNTEREXAMPLE

Q67 · Hyperparameter-tuning

QUESTION	Q67	MODEL	GPT-5 high · run 17
TOPIC	Hyperparameter-tuning	NEGATIVE MODE	`algorithm_perturbation`
SOURCE	`dp-judge_gpt-5-high_cot_17_responses.json`		
EVIDENCE	Counterexample		

Typeset question (from `cate_2.json`)

Let $\mathrm{TNB}_{\eta,\gamma}$ denote the truncated Negative Binomial distribution with parameters $\eta \in (-1,\infty)$ and $\gamma \in (0,1)$, supported on positive integers. For $\eta \neq 0$, $\Pr[K=k] = \frac{(1-\gamma)^k}{\gamma^{-\eta}-1}\prod_{\ell=0}^{k-1}\frac{\ell+\eta}{\ell+1}$ and $\mathbb{E}[K] = \frac{\eta(1-\gamma)}{\gamma(1-\gamma^{\eta})}$. For $\eta = 0$, $\Pr[K=k] = \frac{(1-\gamma)^k}{k\log(1/\gamma)}$ and $\mathbb{E}[K] = \frac{1/\gamma-1}{\log(1/\gamma)}$. If $\mathcal{D} = \mathrm{TNB}_{0,\gamma}$ and each M_j satisfies (α,ε)-RDP for all $\alpha > 1$, the algorithm satisfies (α,ε')-RDP with $\varepsilon' = (2-\frac{1}{\alpha})\varepsilon + \frac{\log(1/\gamma)}{\alpha} + \frac{\log(1-\gamma)-\log\gamma-\log(\log(1/\gamma))}{\alpha-1}$.

Algorithm 4: Private Hyperparameter Tuning

Input : Dataset D; DP base training mechanisms $\{M_j\}_{j=1}^{m}$ (one per hyperparameter setting j); distribution for number of runs $\mathcal{D}$; validation metric score : model × dataset → $\mathbb{R}_+$
Output: Best model $\hat{\theta}$ and chosen hyperparameter $j^\star$

Draw $K \sim \mathcal{D}$
Split D into D_{train} and D_{test}
for $k \leftarrow 1, 2, \ldots, K$ do
  Sample $j_k \sim \mathrm{Unif}(\{1,\ldots,m\})$
  $\theta_k \leftarrow M_{j_k}(D_{train})$
  $s_k \leftarrow \mathrm{score}(\theta_k, D_{test})$
$k^\star \leftarrow \arg\max_{k\in\{1,\ldots,K\}} s_k$
$j^\star \leftarrow j_{k^\star}, \quad \hat{\theta} \leftarrow \theta_{k^\star}$
return $(\hat{\theta}, j^\star)$

Selected model response (wording preserved; mathematics typeset)

Short-answer intuition. As written, the algorithm uses the private test split D_{test} to choose which model to output, without any DP noise in the scoring step. This alone means the algorithm is not DP with respect to the whole dataset D, regardless of how K is randomized or how private the M_j are on D_{train}.
A simple counterexample shows this: take each M_j to ignore the data and output a fixed model (hence 0-RDP), and design score so that the arg max deterministically reveals a bit of D_{test}. The overall algorithm is then not private at all. Thus the stated (α,ε')-RDP guarantee cannot hold for the algorithm as written.

[...]

Conclusion. The claim as stated is not correct for the provided algorithm.

No

FLAG · VALID COUNTEREXAMPLE

"Take M_j to ignore the data and output a fixed model (hence 0-RDP), and design score so that the arg max deterministically reveals a bit of D_{test}; the overall algorithm is then not private at all."

COMMENTS FROM DATASET

The validation score is computed on the private dataset D_{test}, and the arg max is taken directly afterward. Therefore, the algorithm is not differentially private.

EXAMPLE 5 · COUNTEREXAMPLE

Q120 · Sequential or Adaptive Composition

QUESTION	Q120	MODEL	GPT-5 high · run 42
TOPIC	Sequential or Adaptive Composition	NEGATIVE MODE	`conclusion_perturbation`
SOURCE	`dp-judge_gpt-5-high_cot_42_responses.json`		
EVIDENCE	Counterexample		

Typeset question (from `cate_2.json`)

Let $\{\mathcal{M}_i\}_{i=1}^n$ be mechanisms, each satisfying ε-DP under the same neighboring relation. Consider their sequential (possibly adaptive) composition on the same dataset. For any order $\alpha > 1$, does the composed mechanism $\mathcal{M}$ satisfy $(\alpha, n\varepsilon^2/2)$-RDP?

Selected model response (wording preserved; mathematics typeset)

- RDP composes additively: if mechanisms $\mathcal{M}_i$ each satisfy (α, ρ_i)-RDP, their adaptive composition satisfies $(\alpha, \sum_i \rho_i)$-RDP.
- The proposed universal bound is missing its dependence on α.
- Concrete counterexample ($n = 1$). Take randomized response with $\varepsilon = 1$. At $\alpha = 2$,

$$D_2 = \log\left(\frac{p^2}{1-p} + \frac{(1-p)^2}{p}\right), \qquad p = \frac{e^\varepsilon}{1+e^\varepsilon} \approx 0.7311.$$

Therefore

$$D_2 \approx 0.734 > 0.5 = \frac{\varepsilon^2}{2}.$$

Even one ε-DP mechanism need not satisfy $(2, \varepsilon^2/2)$-RDP.

$$\boxed{\text{No}}$$

FLAG · VALID COUNTEREXAMPLE

"Concrete counterexample ($n = 1$): take randomized response with $\varepsilon = 1$. For $\alpha = 2$, $D_2 \approx 0.734 > 0.5 = \varepsilon^2/2$."

COMMENTS FROM DATASET

Consider the following counterexample. Let $\mathcal{M}$ be randomized response on a single bit, which outputs the true bit with probability $p = \frac{e^{0.5}}{1+e^{0.5}}$. Then $\mathcal{M}$ is 0.5-DP. However, the statement in question would imply that $\mathcal{M}$ satisfies $(\infty, 0.125)$-RDP. Since (∞, δ)-RDP is equivalent to δ-DP (because $D_\infty(P\|Q) = \log\sup_y \frac{P(y)}{Q(y)}$), this would further imply that $\mathcal{M}$ is 0.125-DP, contradicting the fact that the privacy loss of randomized response here is exactly 0.5.

EXAMPLE 6 · COUNTEREXAMPLE

Q128 · Smooth_sensitivity

QUESTION	Q128	MODEL	GPT-5 high · run 89
TOPIC	Smooth_sensitivity	NEGATIVE MODE	algorithm_perturbation
SOURCE	dp-judge_gpt-5-high_cot_89_responses.json		
EVIDENCE	Counterexample		

Typeset question (from `cate_2.json`)

Does the following algorithm satisfy 8-zCDP under the replace-one neighbouring relation?

Algorithm 5: Trimmed Mean estimation

Input : Input scalar dataset $D := \{x_1, \dots, x_n\} \in [0,1]^n$, integer $m \in [0, n/2)$
Denote the i-th order statistics of D by $x_{(i)}$, where $x_{(1)} \leq \dots x_{(n)}$
Let $T = \Pi_{[0,1]}\left(\frac{\sum_{i=m+1}^{n-m} x_{(i)}}{n-2m}\right)$, where Π denotes projection operator
Sample $a \sim \mathcal{N}(0,1)$
Get scale $c = \max\{\frac{x_{(n)}-x_{(m)}}{n-2m}, \exp(-m)\}$
return $T + ca$

Selected model response (wording preserved; mathematics typeset)

Short-answer idea. The mechanism adds Gaussian noise with a data-dependent standard deviation $c(D)$. Here $c(D)$ can differ sharply across neighboring datasets, which can make Rényi divergence infinite.
Key derivation (counterexample). Let $n = 2m + 1$ with $m \geq 1$, and consider replace-one neighbors

$$D = (0, 0, \dots, 0), \qquad D' = (0, 0, \dots, 0, 1).$$

The trimmed mean retains only the $(m+1)$-st order statistic, so $T(D) = T(D') = 0$. For the scale,

$$c(D) = \max\{0, e^{-m}\} = e^{-m}, \qquad c(D') = \max\{1, e^{-m}\} = 1.$$

Hence

$$\mathcal{M}(D) \sim \mathcal{N}(0, e^{-2m}), \qquad \mathcal{M}(D') \sim \mathcal{N}(0, 1).$$

For $P = \mathcal{N}(\mu_0, \sigma_0^2)$ and $Q = \mathcal{N}(\mu_1, \sigma_1^2)$, the order-2 divergence $D_2(P\|Q)$ is finite if and only if

$$\sigma_0^2 < 2\sigma_1^2.$$

Apply this with $P = \mathcal{M}(D')$ and $Q = \mathcal{M}(D)$. Since

$$\sigma_0^2 = 1, \qquad \sigma_1^2 = e^{-2m}, \qquad 1 \geq 2e^{-2m} \quad (m \geq 1),$$

we obtain

$$D_2(\mathcal{M}(D')\|\mathcal{M}(D)) = \infty.$$

Therefore no finite ρ can satisfy ρ-zCDP; in particular, the mechanism is not 8-zCDP.

No

FLAG · VALID COUNTEREXAMPLE

"Let $n = 2m + 1$ with $m \geq 1$. Compare $D = (0, \ldots, 0)$ with the neighbor D' having one final 1. Then $\mathcal{M}(D) \sim \mathcal{N}(0, e^{-2m})$, $\mathcal{M}(D') \sim \mathcal{N}(0, 1)$, and $D_2(\mathcal{M}(D') \| \mathcal{M}(D)) = \infty$."

COMMENTS FROM DATASET

Take $n = 3$ and $m = 1$. Then $T = x_{(2)}$ and $c = \max\{x_{(3)} - x_{(1)}, e^{-1}\}$. Consider replace-one neighbors $D = (0, 0, 1)$ and $D' = (0, 0, 0)$. For D, $T = 0$ and $c = 1$, hence $\mathcal{M}(D) \sim \mathcal{N}(0, 1)$. For D', $T = 0$ and $c = e^{-1}$, hence $\mathcal{M}(D') \sim \mathcal{N}(0, e^{-2})$. For $P = \mathcal{N}(0, \sigma_1^2)$ and $Q = \mathcal{N}(0, \sigma_2^2)$ with $\sigma_1^2 > \sigma_2^2$, one has $D_\alpha(P \| Q) < \infty$ iff $\alpha < \sigma_1^2/(\sigma_1^2 - \sigma_2^2)$. Plugging in $\sigma_1^2 = 1$ and $\sigma_2^2 = e^{-2}$ yields the threshold $\alpha \geq 1/(1 - e^{-2}) \approx 1.157$, so in particular $D_2(\mathcal{M}(D) \,\|\, \mathcal{M}(D')) = \infty$. Therefore the mechanism is not ρ-zCDP for any finite ρ (hence not 8-zCDP).

Interpretation and limits

The evidence supports the capability claim: evaluated models sometimes supply exactly the missing proof object - either a neighboring pair/parameter setting that falsifies the guarantee, or a localized change that restores the relevant sensitivity or domain condition.

What this report supports	Existence of strong, checkable “No” explanations across multiple model families and failure modes.
What it does not support	A pass rate, ranking, or claim that every negative response is valid.
Recommended scoring rule	Credit only when the construction satisfies every premise and actually violates the claimed privacy bound, or when the repair is both sufficient and localized.
Common failure to reject	A response that merely says the proof no longer applies, treats a loose sufficient bound as necessary, or proposes a counterexample outside the parameter domain.

Conclusion: Yes. When answering “No,” the evaluated models can provide valid counterexamples or identify repairs. These six examples make that capability visible and auditable, while the scoring caveats prevent a correct label from being mistaken for a correct justification.